Angled launches

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use components to analyze an angled projectile.

SUCCESS CRITERIA - I can...

1. Separate the initial velocity into two components.
2. Use vertical motion to determine flight time.
3. Explain what must be true for your range comparisons to work.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “component” and “projectile” to explain your answer. Sentence starters are optional.

component: the part of a vector along one chosen axis

projectile: an object moving after launch under the stated forces

initial: at the beginning of the interval; subscript i

01 Warm-up: bring back an earlier idea

A downward-velocity fit goes from **0.3 m/s** at **0.01 s** to **3.24 m/s** at **0.31 s**. **Find g** and **explain why fitting an interval** can be better than one noisy point.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $v_{x,\,i} = v_i \cos\theta$; $v_{y,\,i} = v_i \sin\theta$; $t_{flight} = \mathbf{2}\, v_{y,\,i}/g$
range = $v_{x,\,i}\, t_{flight}$.
θ above horizontal; g = **10 m/s²**. Equal launch/landing height; no air resistance.

Worked reasoning: A projectile has initial components **4 m/s** horizontal and **10 m/s** upward. With g=**10 m/s²** it reaches the top in **1 s** and returns to launch height in **2 s**; range=**8 m**.

Your turn: Sketch its path and velocity arrows at launch, top and return.

Sentence starter: In my work, I used … to show … . I labeled it … .



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

At the highest point, **which velocity** component is zero and which remains? Is acceleration zero?

Sentence starter: I notice … . This supports … because … .



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

04 Apply the physics

A ball launches from level ground at **20 m/s**, **30°** above horizontal, and lands at the same height.
1. Use g = **10 m/s²** and ignore air resistance.
2. **Draw the initial velocity components**.
3. **Find flight time**, range and maximum height.

Sentence starter: I found … [unit]. This value means … in this situation.



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

In this ideal projectile calculation, why can you double the time to the top only when the object lands at its launch height?

Word help: ideal = using stated simplifying conditions

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A simplified calculation predicts position (**-70**, **90**) km with bounds **±7 km** in x and **±9 km** in y. **Write a short briefing giving** both intervals and two reasons it is not a real asteroid collision forecast.
Label the given and predicted quantities on the diagram.

Sentence starter: I predict … . I used … to make this prediction because … .

y position (km)



x position (km)

Marzano check - circle: 1 2 3

07 Exit check

For speed **20 m/s**, compare ranges at **30°** and **60°** on level ground. **Explain which has longer flight** time. **Label the diagram** with the known quantities before calculating. Ignore air resistance; use g = **10 m/s²** when needed.

Sentence starter: Both … . They differ in … because … .

Separate horizontal and vertical motion

Label x and y; state positive directions.

Velocity:

Acceleration / force:

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

For a launch at **20 m/s** from level ground, find the two angles giving an ideal range of **20 m** with g=**10 m/s²**. **Compare flight times**. **Label the diagram** with the known quantities before calculating. Ignore air resistance; use g = **10 m/s²** when needed.

Word help: ideal = using stated simplifying conditions

Sentence starter: Both … . They differ in … because … .



Add velocity / acceleration arrows.

Marzano check - circle: 1 2 3